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Section 1

01. Business Analyst - is the art of turning numbers into decisions. It is about asking a business questions like "why are we loosing customers?", should we open a new warehouse?" and using the data to answer those questions.

- Example - Microsoft: After moving an engineering team across offices, used workplace data to ensure whether to bringing people closer together actually made them collaborate more.
- Source - Harvard Business School online.

02. Data Analyst - it is the broader craft of working with data, collecting it and cleaning it. It's less about "what should our company do" more about "what does this data actually tell us?"

- Example - Massachusetts General hospital: uses data analytics to identify patients most likely to be readmitted after discharge. By flagging in high risk ones early.
- Source - Coherent Solutions.

03. Business intelligence (BI) - is the reporting layer of a business the dashboard, charts and weekly summaries that tell you how things are going. It takes data from all over the company and packages it up so people can actually see it.

- Example - Netflix: uses BI tools so products and business teams can dig into viewing data themselves no waiting for a data team to pull a report.
- Source - Accelodata

04. Data-driven decision making - is the commitment, culture not just technical to making decisions based on evidence rather than instict, tradition or whoevers the loudest in the room.

- Examples: Industry-wide (micro-strategy survey) 57% of global enterprise now have a Cheif Data Officer - a sign that DDM has moved from tech project to a boardroom priority.
- Source - Databox

05. Big data - is what happens when your data gets so large, so fast-moving or so messy that a normal spreadsheet or database falls over trying to handle it. Handling big data is an engineering problem before it's an analytics problem

- Example - Netflix is capturing what every single person watches, skips, pauses, searches for and abandons
- Source - Mentionlytics

06. Descriptive analytics is the foundation, it tells you what already happened by summarising your historical data. Every KPI dashboard you've ever seen, every 'monthly performance report' every chart showing revenue over time all this is descriptive analytics.

- Example - 'Sales were down 15% in Q3' - that's descriptive it's a starting point not the answer.
- Source - AWS.

07. Diagnostic analytics is detective work. You've seen it in a crime scene and now your figuring out who did it and why. It involves drilling into data, comparing cohorts to find factors behind results.

- Example - Supply chain scenario, A warehouse notices a Q3 stock out problem. Diagnostic analytics traces it back to a supplier delay that coincided with a promotional campaign.
- Source - AWS.

08. Predictive analytics uses patterns from historical data to make educated guesses about the future. Not certainties or probabilities. It uses statistical models and machine learning to find signals in past behaviour and project them forward.

- Example - Massachusetts General Hospital predicted which patients were at high risk of readmission after discharge. Proactive intervention based on those predictions cut readmissions by 22%.

09. ~~D~~ Prescriptive analytics is the top of the pyramid and the rarest in the practice. It doesn't just predict what will happen it tells you about it.

• Example - UPS uses prescriptive routing models that factor in traffic, weather, package weight and outputs the specific route each driver should take.

10. Data literacy is the ability to read data critically understand what it does and doesn't say and communicate with it confidently. It doesnt mean knowing how to code or build models.

• Example - Industry wide 57% of global enterprises now have Cheif Data Officer partly because organisations realise that analytics tools are useless if people making decisions can't interprete the outputs.

Section 2

11. Structured data is data that lives neatly in a table, rows & columns, clearly labelled, easy to sort and filter. Every spreadsheet you've every opened is structured data.

• Example - Retail :- One row per purchase, columns for Customer ID, store location.

12. Unstructured data is everything that doesn't fit in a neat table. A customer complaint email. A product review. A call centre recording. And yet this is where most of the world's information actually lives some estimated put it at 80-90% of all data generated

Example - Netflix, every thumbnail, every user review, every second of video content is unstructured.

13. Data warehouse is a central place where a company brings together data from all its different systems the CRM, e-commerce platform, it cleans it up and stores it in a format that's built for analysis.

Example - Chipotle :- Before centralising into a data warehouse, different reasturants teams had different views of data - no one could see the full national picture.

14. Data mining is the process of digging through large data sets to find patterns, relationships and insights that weren't obvious going in. You're not testing a specific hypothesis you exploring.

• Example · Amazon's customers who bought this also bought' recommendation engine is data mining at scale finding purchase co-occurence patterns across hundres of millions of transactions

15. Data visualization is the practice of turning numbers into pictures, charts, maps that make patterns visible at glance. The human brain process visuals for faster than tables of numbers

• Example - Financial reporting: A line chart showing revenue over 24 months communicates trend, seasonality and the impact of a specific event far more clearly than a table of 24 numbers

16. Dashboard is a single screen that brings togethe the most important metrics for a team, function or business updated regularly so people can see at a glance whether things are on track, ~~A good~~

• Example - Chipotle: Operations managers can see a live view of perfomance across all resturants nationally speed of service waste levels, order volume without waiting for a weekly report or calling individual locations

17. Metric - A metric is simply any number you track. If you can measure it and write it down, it's a metric.

• Example - E-commerce site: "Number of pages viewed per session' is a metric, It gets tracked. It shows up in analyti reports.

18. Key Perfomance indicator (KPI) - is a metric that has been promoted. It's not just tracked it's monitored against a target, reviewed in leadership meeting and used to decide whether to act. It directly reflects progress toward a strategic goal

• Example: A logistic company - On-time delivery rate is a KPI. It's on the CEO's dashboard. It falls below 95% something changes staffing, routing, supplier contracts.

19. Benchmark is the number you compare your performance against. Without a bench, a metric tells you nothing. A 3% conversion rate sounded bad until you find out the industry average is 1.5% and you're actually leading your category.

- Example Email marketing - an open rate of 22% means nothing on its own, Benchmarked against the industry average for your sector. It tells you whether your ahead, behind or average.

20. Segmentation is the practise of splitting your customers into groups that share meaningful characteristics - so you can treat each group differently. Instead of sending the same email to everyone.

- Example - Netflix doesn't show everyone the same homepage. It segments viewers by watching history, genre preference, time of the day they watch.

Section 3

21. Conversation rate ~~is the percentage of visi~~ a key performance indicator that measures the proportion of website or campaign visitors who ~~delete~~ complete a desired goal.

- Example - an e-commerce dashboard shows a 23% conversion rate, meaning 23 out of every 100 visitors complete a purchase

~~22~~ 23. Customer lifetime value - A predictive metric estimating the net profit attributed to the entire future relationship with a customer. Often modeled using historical purchase data and retention patterns.

- Example - A subscription service calculate CLV ~~£~~ at 25 per customer, helping analysts decide how much they can spend on acquisition without losing profit.

~~23~~ 22. Customer acquisition cost - a cost efficient metric that divides total marketing and sales spend by the number of new customers acquired.

- Example - A startups campaign report shows a CAC of 24 meaning each new customer cost 24 in advertising and sales expenses.

24. Chuun rate - a retention metric that measures the percentage of customers lost during a specific time frame. It is critical for forecasting revenue stability.

- Example - A streaming platform analytics dashboard shows a churn rate of 26% signalling that more than a quater of subscribers cancel monthly

25. Customer retention rate - A longtudinal metric that tracks the percentage of customers who remain active over a given period.

- Example - A Saas company reports 78% retention rate quaterly analytics, showing that most users continue renewing their subscriptions.

26. Net promoter score - a customer expirience metric derived from survey data, measuring the likelihood of customers recommending a product or service.

- Example - A hotels chain analytics report shows an NPS of 28, indicating moderate customer advocacy and room for improvement in service quality.

27. Average order value - a revenue metric that calculates the mean value of transactions over a set period. Often used to assess upselling and cross-selling strategies.

- Example - An online retailers analytics dashboard shows an AOV of 22 meaning each transtion generates 22 revenue in average

28. Sales funnel - A process visualization that maps customer progression from awareness to purchase, often tracked with conversation metrics at each stage

- Example - A marketing analytics dashboard shows drop offs rates at each funnel stage awarness helping optimize campaigns

Section 4

29. Marketing campaign - A coordinated set of

marketing activities designed to acheive a specific business goal

• Example - Coca colas "share a coke" campaign replaced logos with names to drive engagement and sales.

30. Post-campaign analysis - evaluation stage where analysts measure campaign performance against objectives using KPIs like ROI

• Example - After Nike runs social media campaign then analyze which channels drove the most conversations and whether sales increased.

31. Incremental revenue - the additional revenue generated directly because the campaign, beyond what would have happened without it.

• Example - If a retailer usually makes 100k monthly but a holiday campaign boosts sales to 130k the increment revenue is 30k.

32. Return on Investment - the profitability test calculated as net profit divided by campaign costs

• Example - A company spends 10,000 on ads and earns 40,000 in profit.

33. Return on Ad spend - A marketing specific efficieny metric showing revenue earned per advertising dollar spent.

Example - If amazon spends 1,000 on google ads and earn 5,000 on sales ROAS = 5:1

34. Click - through sales - the percentage of people who click on an ad or link after seeing it.

• Example - If 10,000 people see a facebook ad and 340 clicks CTR = 3,4%. Analysts use CTR to judge ad relevance.

35. Bounce rate - The percentage of visitors who leave a site quickly withought engaging

Example - If 1,000 people land on a blog page and 350 leave immediately, bounce rate = 35%.

36. A/B testing - An expirimental method comparing two versions of a campaign element to see which performs better.
Example - Spotify tests two email subjects lines to see which gets more opens.

37. Lead generation - process of capturing potential customers information for future nurturing
• Example : HubSpot offers free eBooks in exchange for email address, generating leads for their sales team

38. Marketing attribution - the analytical process of determining which channel or touch point deserves credit for a conversion
• Example - If a customer sees google ad, reads a blog and then clicks an email before buying, attribution model decide which channel gets the credit.

Sources

1. Harvard business law
2. Mckinsey
3. Investopia
4. Statista
5. course textbook.